



Designing and Building AI Products and Services

Module 2 Office Hours
Bruce Lam



Agenda

01 **Machine Learning**
What is it?

02 **Prediction vs.
Classification**

03 **Churn Prediction Case**

04 **Unsupervised Learning**

05 **ML or AI?**

Terms



model (n)

[ˈmɒdəl] / noun

1. a mathematical representation of a real world process; a predictive model forecasts a future outcome based on past behaviors.



algorithm (n)

[ˈælgəˌrɪðəm] / noun

1. a set of rules used to make a calculation or solve a problem.



training (v)

[ˈtreɪnɪŋ] / verb

1. the process of creating a model from the training data. The data is fed into the training algorithm, which learns a representation for the problem, and produces a model. Also called "learning".



regression (n)

[rɪˈɡreʃən] / noun

1. a prediction method whose output is a real number, that is, a value that represents a quantity along a line. Example: predicting the temperature of an engine or the revenue of a company.



classification (n)

[ˌklæsəfəˈkeɪʃən] / noun

1. a prediction method that assigns each data point to a predefined category, e.g., a type of operating system.



target (n)

[ˈtɑːɡət] / noun

1. in statistics, it is called the dependent variable; it is the output of the model or the variable you wish to predict.



training set (n)

[ˈtreɪnɪŋ set] / noun

1. a dataset used to find potentially predictive relationships that will be used to create a model.



test set (n)

[test set] / noun

1. a dataset, separate from the training set but with the same structure, used to measure and benchmark the performance of various models.



feature (n)

[ˈfi:tʃər] / noun

1. also known as an independent variable or a predictor variable, a feature is an observable quantity, recorded and used by a prediction model. You can also engineer features by combining them or adding new information to them.



overfitting (v)

[ˌoʊvərˈfɪtɪŋ] / verb

1. a situation in which a model that is too complex for the data has been trained to predict the target. This leads to an overly specialized model, which makes predictions that do not reflect the reality of the underlying relationship between the features and target.



Term	Similar terms
Observations	Data points
	Samples
	Instances
	Records
	Rows
Features	Independent variable
	Explanatory variable
	Attributes
	Columns
Target	Dependent variable
	Outcome variable
	Output
	Class or label (in classification)
Feature space	Vector space
	Embedding space
Data structure	Matrix
	Array
	Vector
	Tensor
	Graph
Penalization	Regularization
Estimation	Prediction
	Learning

https://www.researchgate.net/figure/Different-terminology-used-in-machine-learning-statistics-and-computer-science_tbl3_342795298

<http://www.franks.world.com/2018/08/23/ten-common-data-science-terms-defined/>

via Ricardo Fernandez]

4d • 

Created By: Aishwarya Naresh Reganti



Machine learning (ML) big picture

The simplest explanation of machine learning you'll ever read



Cassie Kozyrkov

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May 24, 2018 · 4 min read

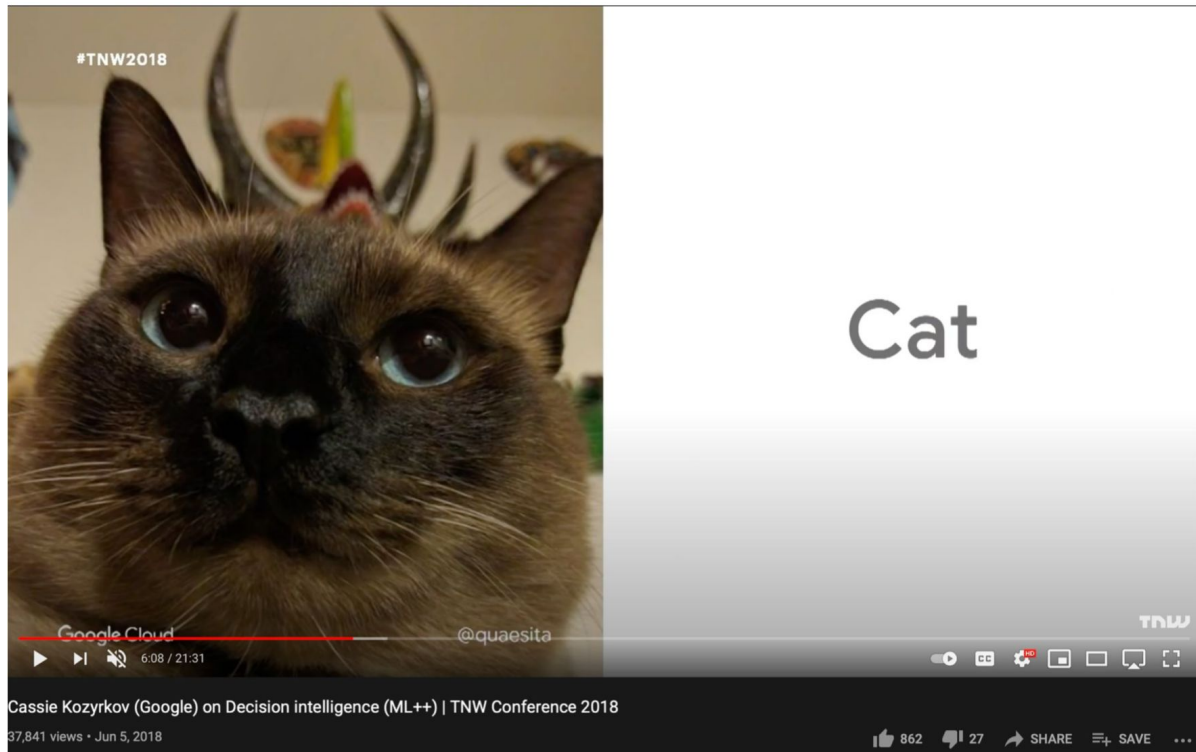


You've probably heard of machine learning and artificial intelligence, but are you sure you know what they are? If you're struggling to make sense of them, you're not alone. There's a lot of buzz that makes it hard to tell what's science and what's science fiction. Starting with the names themselves...

Machine learning is a thing-labeler, essentially.

<https://medium.com/hackernoon/the-simplest-explanation-of-machine-learning-youll-ever-read-bebc0700047c>

Machine learning (ML) big picture



But think about what those instructions would be. What are you actually doing with these pixels? Can you express it? Your brain had the benefit of eons of evolution and now it just works, you don't even know how does it. That recipe is pretty hard to come up with.

Explain with examples, not instructions.

Wouldn't it be better if you could just say to the computer, "Here, look at a bunch of examples of cats, look at a bunch of examples of not-cats, and just figure it out yourself"? *That* is the essence of machine learning. It is a completely different programming paradigm. Now, instead of giving explicit instructions, you program with examples and the machine learning algorithm finds patterns in your data and turns them into those instructions you couldn't write yourself. No more handcrafting of recipes!

AI allows you to automate the ineffable.

Examples of work



IMAGE CREDITS: [KYLE GRILLOT/BLOOMBERG](#) / GETTY IMAGES



Lucas Ropek

Meta will record employees' keystrokes and use it to train its AI models

Meta has found a new source of training data for its AI models: its own employees. The company plans to use data culled from the mouse movements and keystrokes of its own staff in its pursuit to build more capable and efficient artificial intelligence.

The story, which was first [reported by Reuters](#), shows the lengths to which tech companies are going to find new sources of training data — the lifeblood of AI models that helps the programs learn how to more effectively carry out tasks and respond to user queries.

Lecture Recording

Mark Zuckerberg Addresses Staff Ahead of Mass AI Layoffs



that our system needs to be very
good at using computers.

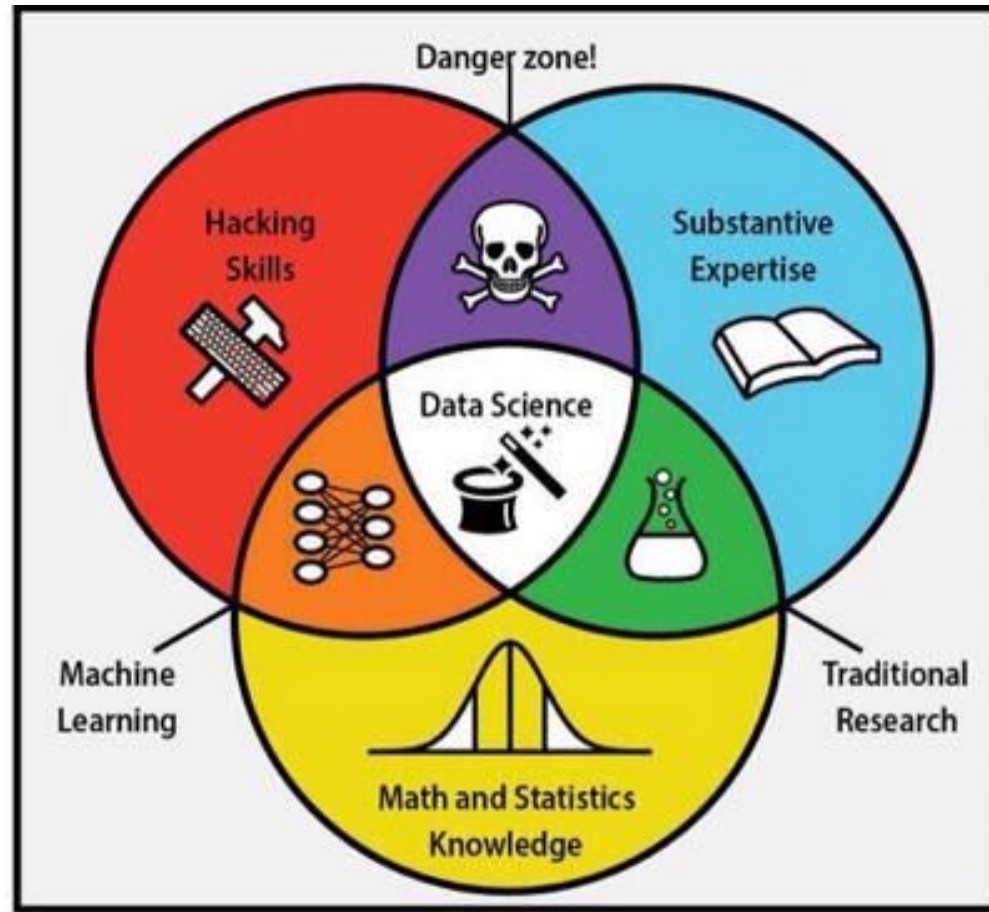
<https://www.youtube.com/watch?v=sdPZrhWdJw&list=WL&index=3>

Meta Pauses Employee-Tracking Program Following Internal Data Leak

The move comes after the company left potentially sensitive data from the initiative exposed internally.

<https://www.wired.com/story/meta-pauses-employee-tracking-program-following-internal-security-breach/>

Why the technical details?



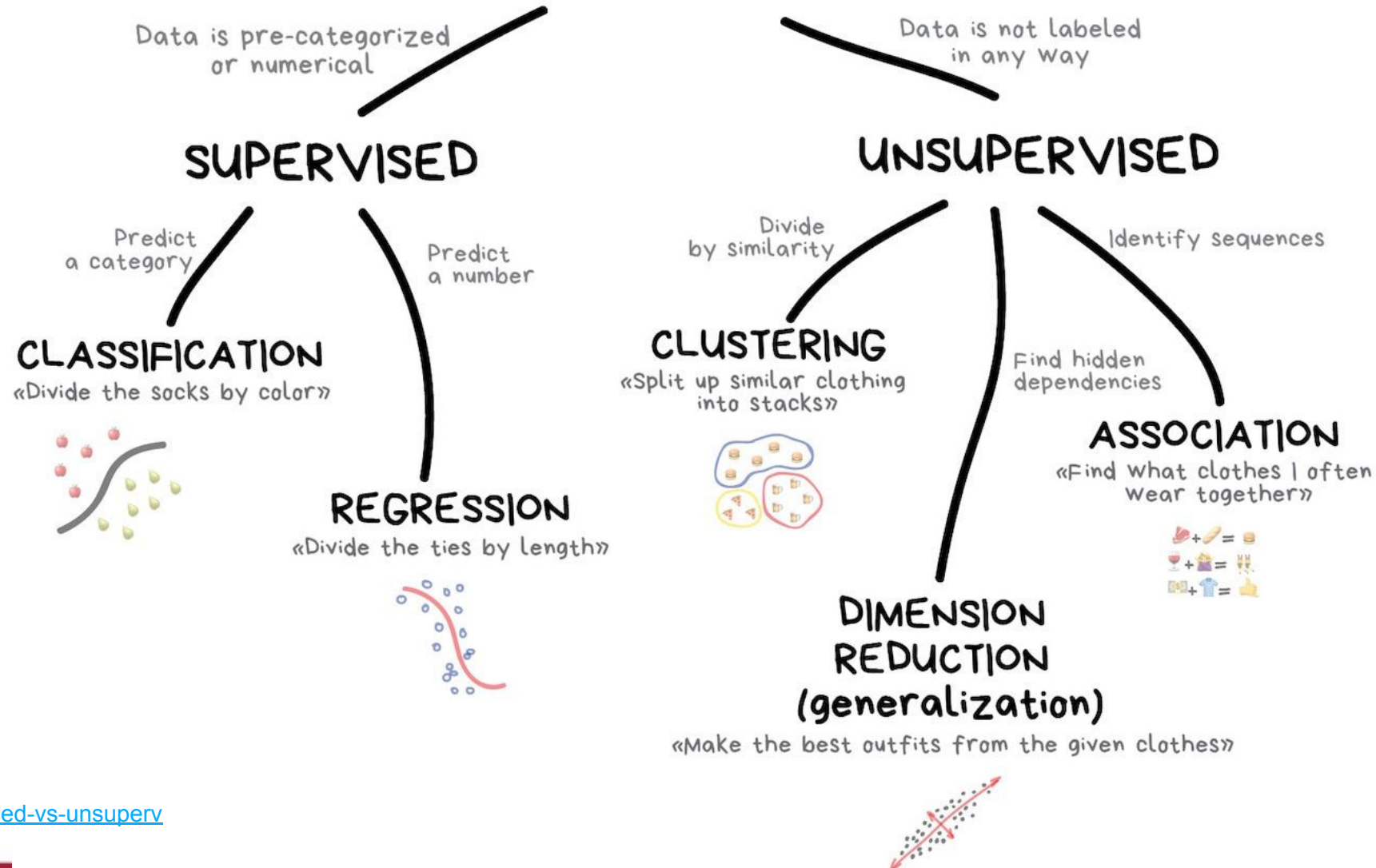
The diagram illustrates the relationship between different levels of Artificial Intelligence (AI) through concentric circles. The outermost circle represents the broadest field, while the innermost circle represents the most specific and advanced sub-field.

- Artificial Intelligence (Outermost Circle):** This level includes the most general concepts and applications of AI, such as:
 - Intelligent Robotics
 - Augmented Programming
 - Algorithm Building
 - Hypothesis Testing
 - Support Vector Machine
 - Dimensionality Reduction
 - Decision Trees
 - Augmented Programming
 - Supervised Machine Learning
 - UnSupervised Machine Learning
 - K Means
 - Reinforcement Learning
 - Speech Recognition
 - AI Ethics
 - Emergent behavior
- Machine Learning (Second Circle):** This level focuses on the core algorithms and techniques used in AI, including:
 - Perceptron
 - Feed Forward
 - Backpropagation
 - Deep Feed Forward
 - Feed Forward
 - Logistic Regression
 - PCA
 - Self organising maps
 - Liquid State Machine
 - Convolution Neural Network(CNN)
 - Long Short Term Memory Network(LSTM)
 - Boltzmann Machine
 - Deep Believe Network
 - Hopefield Network
 - Multi layer perceptron
- Neural Networks (Third Circle):** This level delves into the specific architectures of neural networks, such as:
 - Recurrent Neural Network(RNN)
 - Foundation Model
 - Agents
 - BigGAN
 - Deep Reinforcement Learning
 - GPT
 - BERT
 - Epochs
 - Transfer Learning
 - Large Language Model
 - Langchain
 - Multimodal AI
 - One Shot Learning
 - Few Shot Learning
 - RLHF
 - Hallucination
 - QLoRA
- Deep Learning (Fourth Circle):** This level focuses on the specific models and techniques used in deep learning, including:
 - Generative Adversarial Networks(GANs)
- Generative AI (Innermost Circle):** This level represents the most specific and advanced sub-field of AI, focusing on the creation of new content, such as:
 - Generative Adversarial Networks(GANs)



Unsupervised vs. Supervised

CLASSICAL MACHINE LEARNING



<https://devopedia.org/supervised-vs-unsupervised-learning>

Comparison of Machine Learning Paradigms

Supervised Learning

Strengths:

- Learns complex patterns from labeled data
- Provides clear predictions and classifications
- Achieves high accuracy with representative training data

Weaknesses:

- Requires labeled training data, which can be costly
- May not perform well on tasks with unclear patterns or noisy data
- Can be prone to overfitting

Unsupervised Learning

Strengths:

- Discovers hidden patterns in unlabeled data
- Useful for clustering, dimensionality reduction, and anomaly detection
- Does not require labeled data

Weaknesses:

- Interpretation of discovered patterns can be challenging
- Performance is sensitive to hyperparameters and data structure
- Evaluating results can be difficult without ground truth

Reinforcement Learning

Strengths:

- Learns optimal actions through trial-and-error in an environment
- Suitable for sequential decision-making problems
- Can adapt and learn from feedback in dynamic environments
- Can discover novel and unexpected solutions

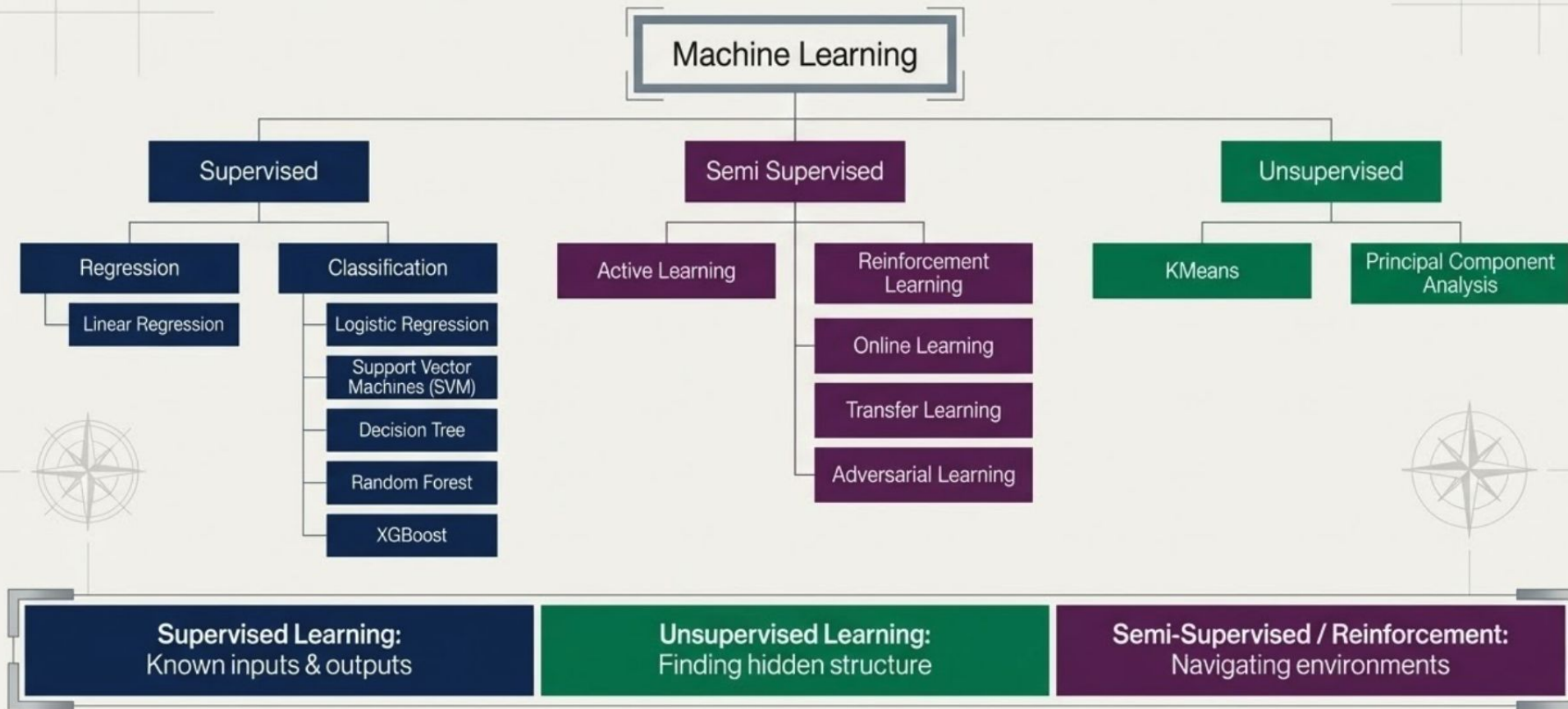
Weaknesses:

- Requires a well-defined reward function and environment
- Computationally intensive and requires many interactions
- Sensitive to the exploration-exploitation trade-off
- Difficult to scale to complex, high-dimensional problems

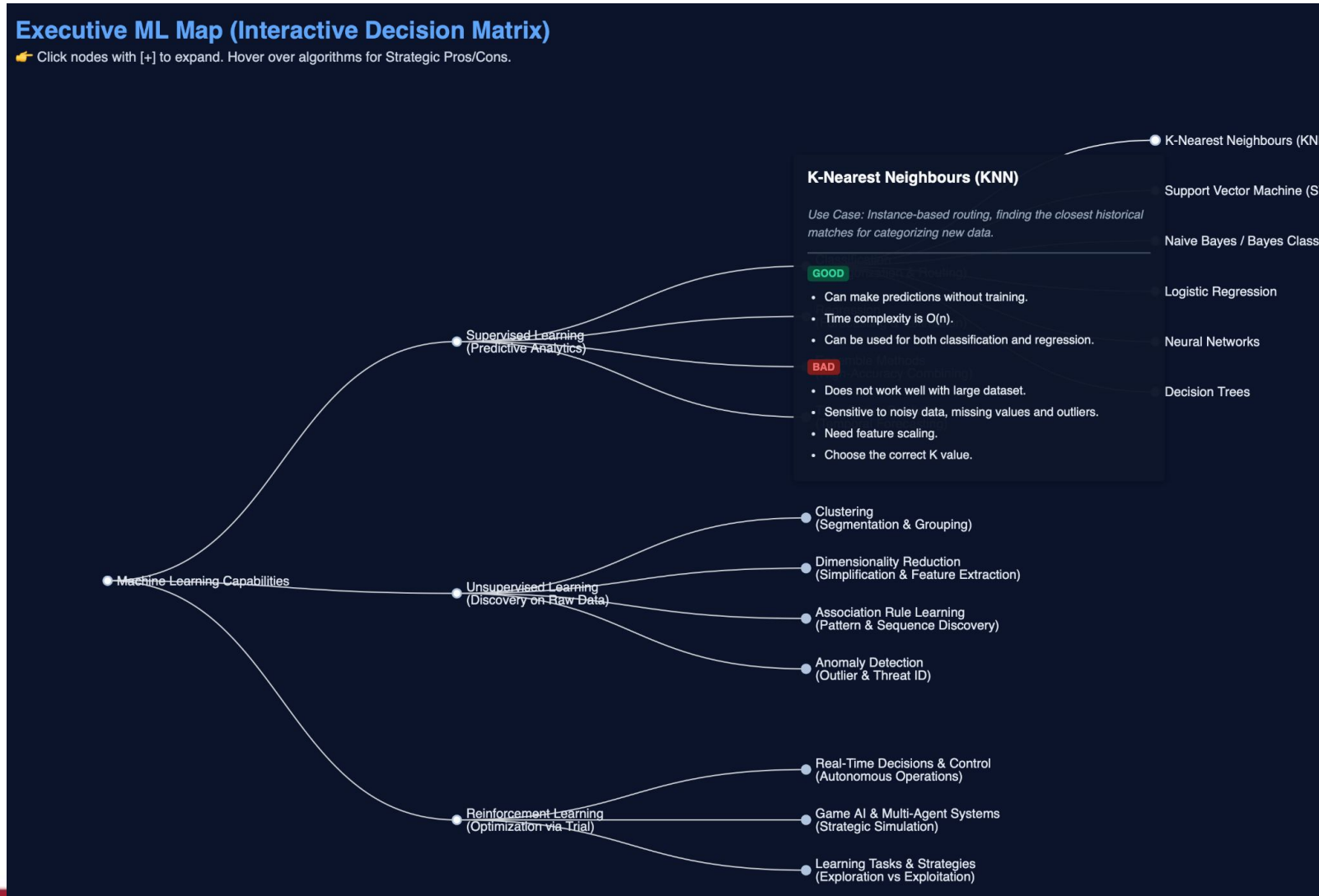
<https://medium.com/analytics-vidhya/overfitting-detection-prevention-848e6adeabf8>

Comparison of Machine Learning Paradigms

The Machine Learning Taxonomy Map



Comparison of Machine Learning Paradigms



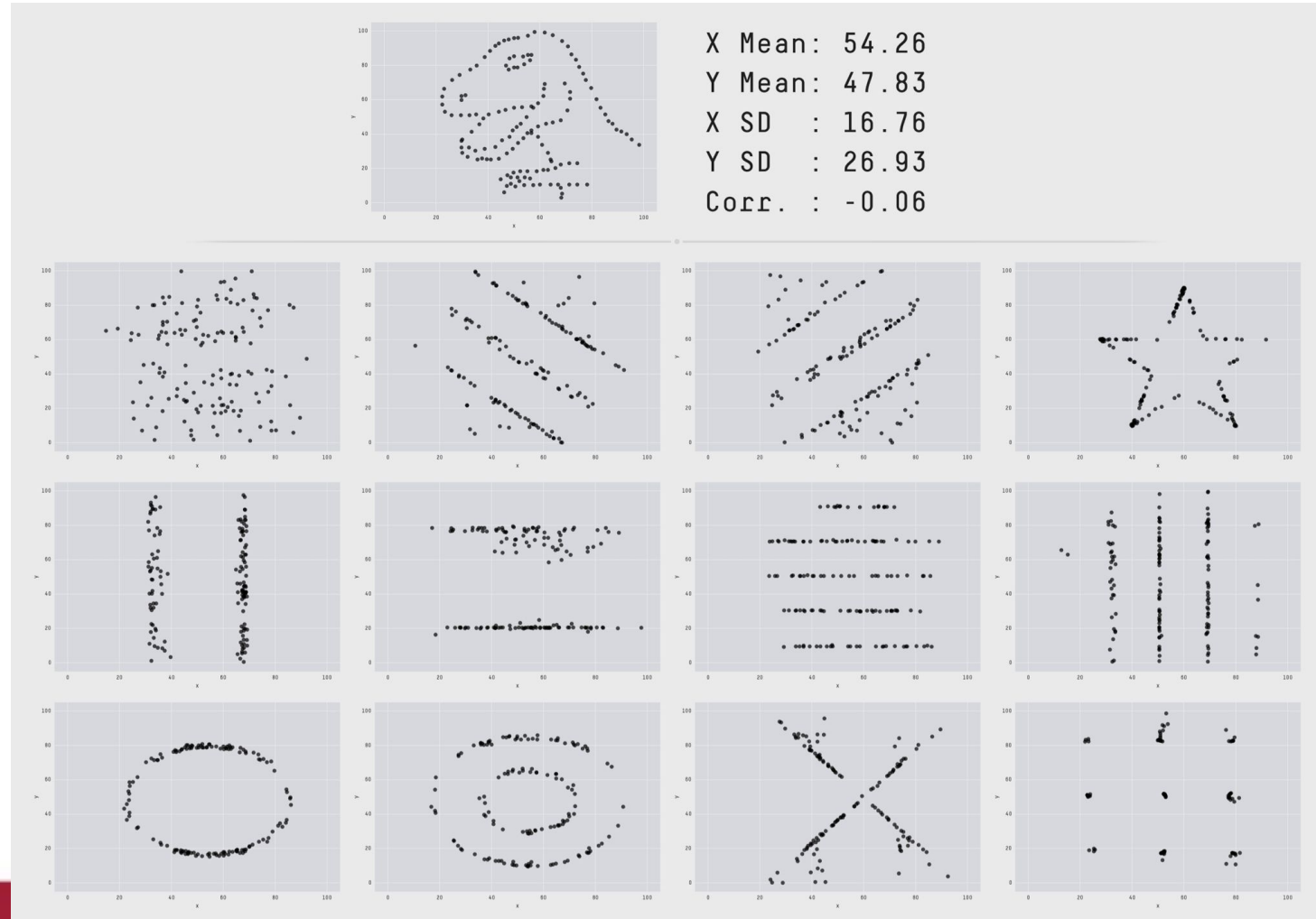
Cheatsheet

Machine Learning Algorithms

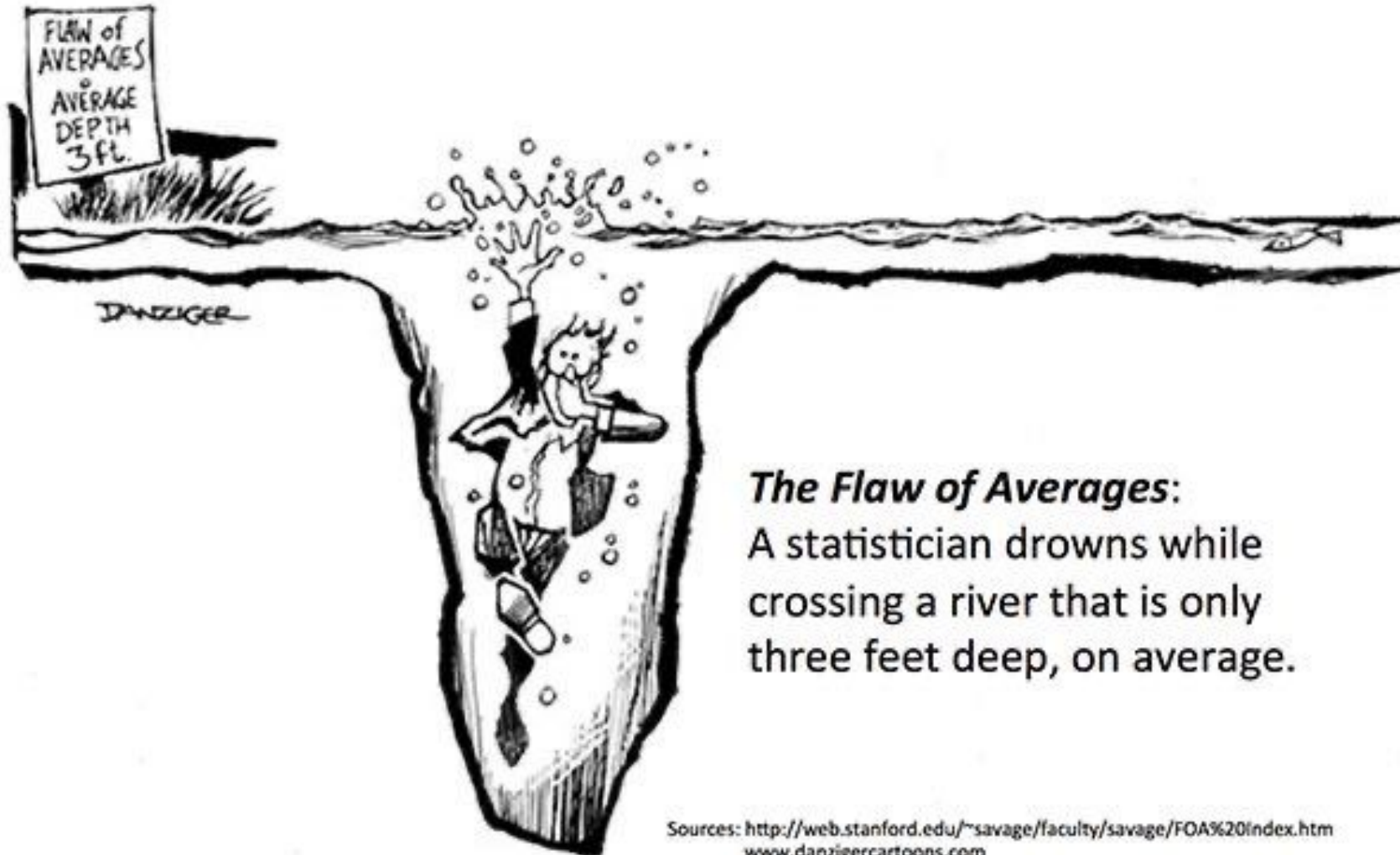
Algorithm	Type	Best Use Case	Key Formula or Logic	Assumptions	Pros	Cons	When Not to Use	Real-World Example
Linear Regression	Supervised	Predicting continuous values	$Y = b_0 + b_1X + B_2X^2 + \dots$	Linearity, independence	Simple, interpretable, fast	Sensitive to outliers, non-linear limits	Data with strong non-linearity	House price prediction
Logistic Regression	Supervised	Binary classification	$P = 1 / (1 + e^{-(b_0 + b_1X + \dots)})$	Log-odds linearity	Probabilistic, interpretable	Weak with non-linear boundaries	Data is highly non-linear	Spam detection
Decision Tree	Supervised	Classification and regression	Recursive binary split	None	Easy to interpret	Overfitting, unstable	Noisy or complex datasets	Loan default prediction
Random Forest	Supervised	Ensemble modeling	Bagging plus averaging trees	Tree independence	High accuracy, robust	Slower, less interpretable	Need real-time results	Fraud detection
Gradient Boosting	Supervised	High-performance modeling	Additive trees minimizing loss	Sequential dependency	State-of-the-art accuracy	Overfitting, needs tuning	When interpretability matters	Credit scoring
SVM	Supervised	Max-margin classification	Maximize margin using kernel trick	Separability, scaling	Works in high dimensions	Slow on large data	Large noisy datasets	Facial recognition
KNN	Supervised	Few-shot classification	Distance-based majority vote	Feature scaling	Simple, no training phase	Slow, noise sensitive	High-dimensional noisy data	Recommender systems
Naive Bayes	Supervised	Text classification	Bayes theorem plus feature independence	Independent features	Fast, good with text data	Fails with correlated features	Feature dependency present	Sentiment analysis
K-Means	Unsupervised	Customer segmentation	Minimize intra-cluster distance	Spherical, equal clusters	Fast, easy to implement	Needs K, sensitive to scale	Non-spherical data	Customer segmentation
Hierarchical Clustering	Unsupervised	Data structure understanding	Nested dendrogram	Distance metric	No need for K, visual	Memory and compute intensive	Very large datasets	Gene expression analysis
PCA	Dimensionality Reduction	Reducing feature dimensionality	Eigenvectors of covariance matrix	Large variance important	Noise reduction, speed-up	Hard to interpret	All features are interpreted	Image compression
Neural Networks (MLP)	Supervised	Complex pattern modeling	Weighted sums plus activation functions	Enough data, scaling	Non-linear learning power	Needs large data and tuning	Small data, low compute	Image classification
CNN	Supervised	Image, video, spatial data	Convolution plus pooling layers	Grid-like spatial data	Excellent for images	High resource demand	Sequence or text data	Self-driving vision
RNN	Supervised	Sequence over time	Feedback loops over time	Sequential structure	Time-series and text ready	Vanishing gradient	Long sequences	Stock prediction
Transformer (BERT, GPT)	Supervised or Self-supervised	NLP tasks, chat, translation	Attention mechanism plus position encoding	Large training data	Long content, fast	Heavy compute, large model	Small projects	ChatGPT, translation tools
Autoencoders	Unsupervised	Compression and anomaly detection	Encoder-decoder plus reconstruction loss	Symmetric network	Effective denoising	Can overfit, black-box	When no compression needed	Fraud detection
DBSCAN	Unsupervised	Arbitrary shape clustering	Density-based region growing	Cluster density	Noise tolerant, shape-flexible	Falls on varying density	Sparse high-dimensional data	Geo-spatial clustering

https://www.linkedin.com/posts/hana-r-31884313a_ml-ai-ai-activity-7449396131186434048-A0Rj/

Why is visualization critical (EDA - Exploratory Data Analysis)?



The flaw of averages

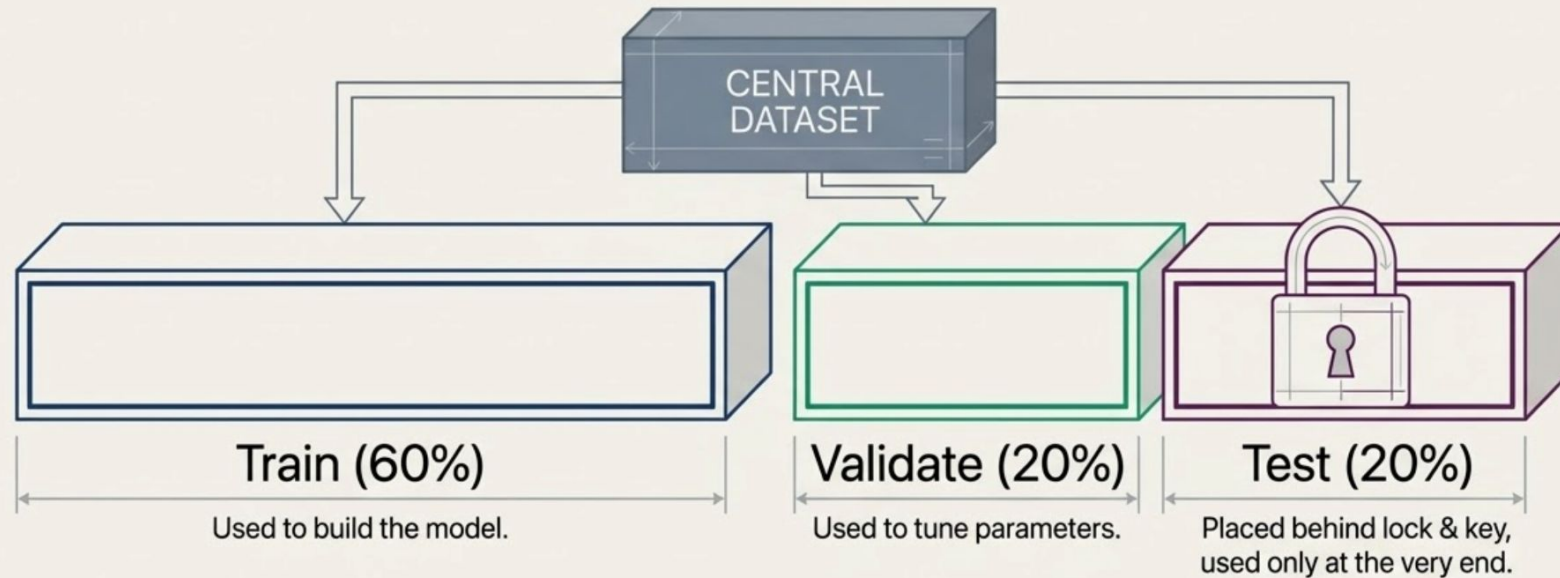


The Flaw of Averages:
A statistician drowns while crossing a river that is only three feet deep, on average.

Sources: <http://web.stanford.edu/~savage/faculty/savage/FOA%20index.htm>
www.danzigercartoons.com

Train/Test Split

The Reality Check: Evaluating Model Generalization



A model that memorizes the training data perfectly has "overfit". For an executive, the only metric that matters is how the model performs on unseen data. The strict **Train/Validate/Test** methodology ensures the algorithm actually learned the underlying pattern, rather than just memorizing the test answers.



Classify vs. Predict



AI, ML and DL use cases

The use cases for AI are incredibly diverse and rapidly evolving, but there are common patterns across industries and verticals. Here is a sampling of possible use cases.

Healthcare and life sciences	Financial services	Government security and defense	Media and entertainment
• Population health • Cancer detection • Chronic illness prediction • Drug discovery • Drug interactions • Genetics • Bioinformatics • Sanitation	• Fraud prevention • Risk management • Investment predictions • Customer service • Digital assistants • Network security automation	• Facial recognition • Video surveillance • Cybersecurity • Satellite imagery • Event prediction • Emergency services • Weather and climate predictions	• Video captioning • Content-based search • Real-time translation • Language processing • Recommendation engines

Manufacturing	Energy	Transportation	Retail
• Smart manufacturing systems • Factory and demand analytics and optimization • Preventive maintenance • Relationship intelligence • Product and service quality	• Wind power generation • Solar forecasts • Oil production optimization • Weather prediction • Demand/consumption predictions	• Autonomous vehicles • Pedestrian and object detection • Lane tracking and traffic patterns • Preventive maintenance • Risk assessment • Logistics	• Supply and demand planning • Predicting buying behavior • Loss prevention • Upsell, cross-sell opportunities • Customer and product movement tracking

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Data Sources

kaggle

Competitions

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Game Arena

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...

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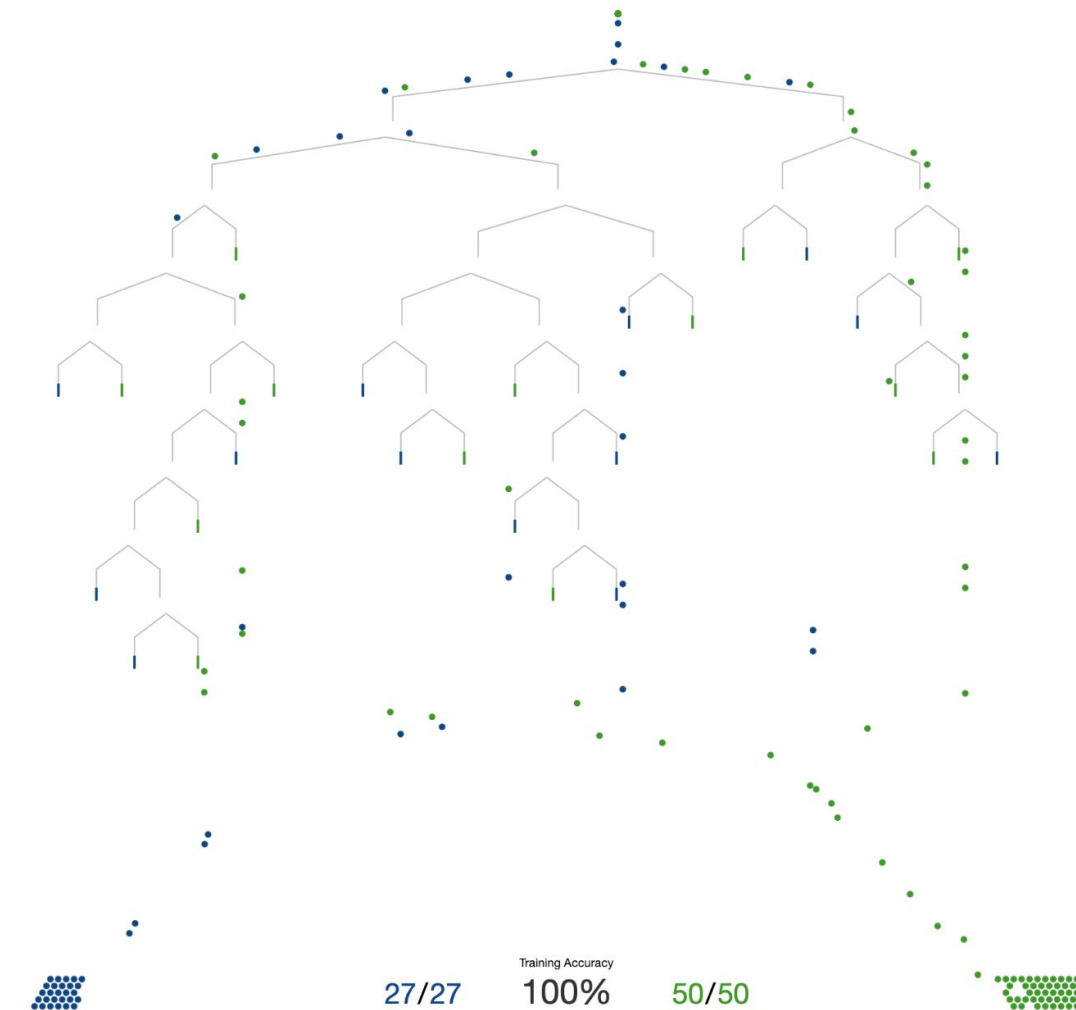
Decision Tree Classification

Making predictions

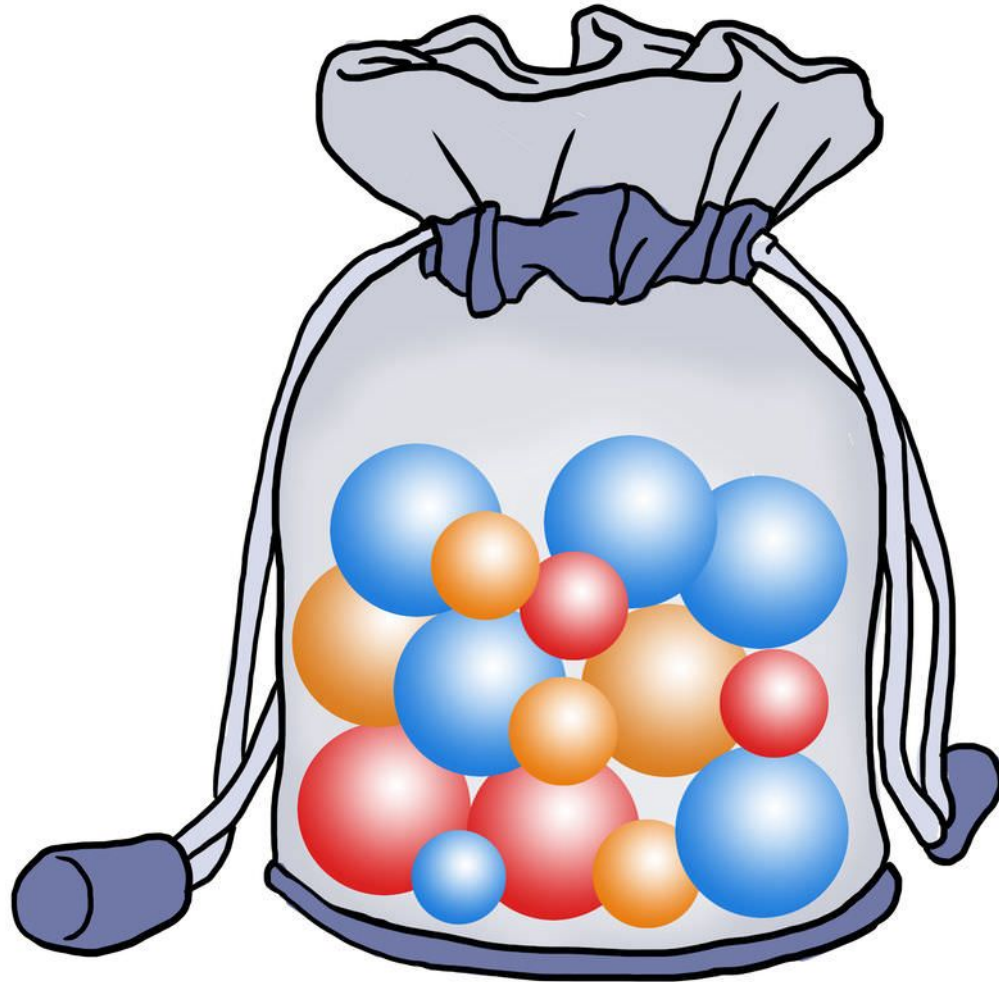
The newly-trained decision tree model determines whether a home is in San Francisco or New York by running each data point through the branches.

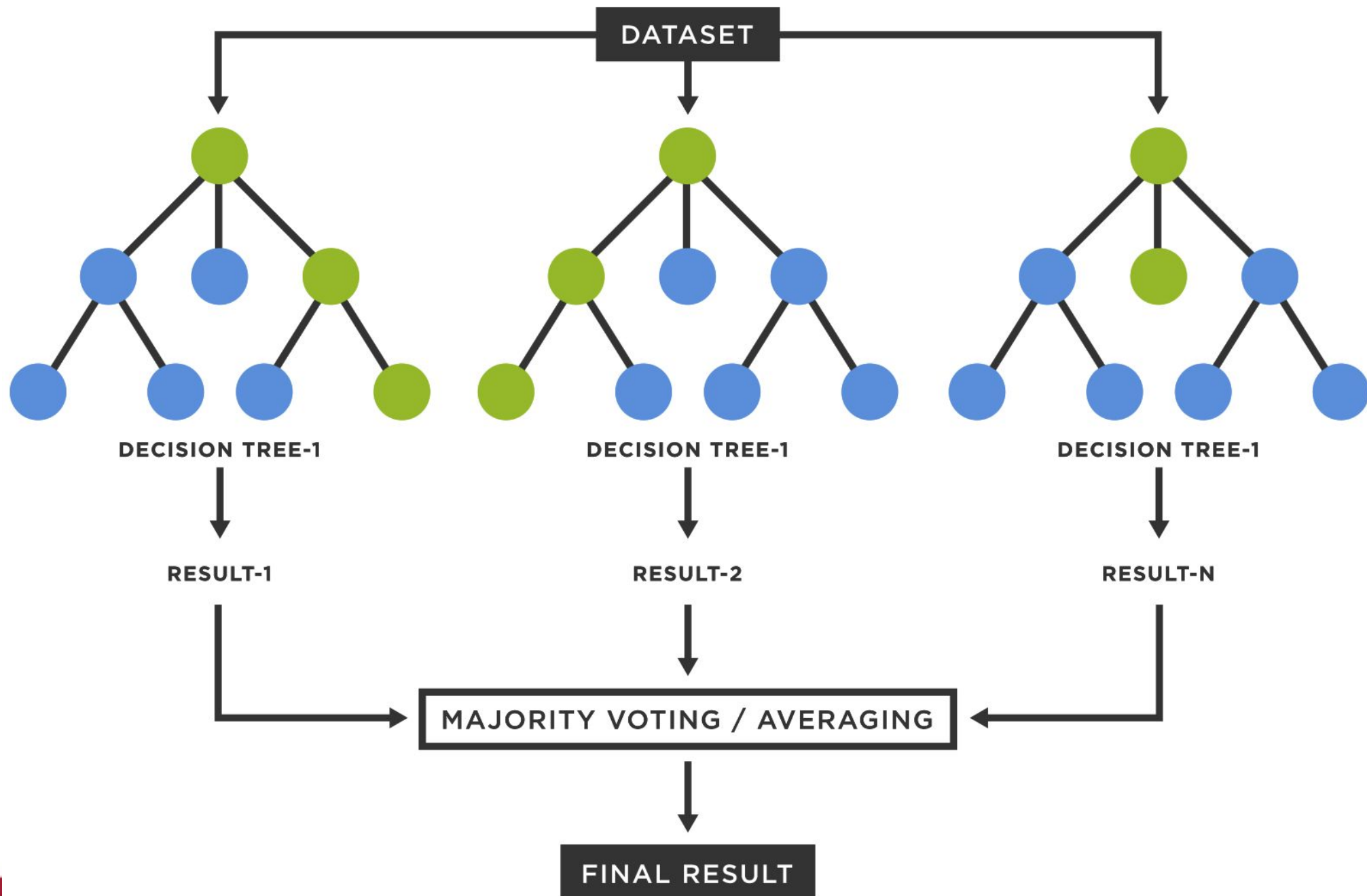
Here you can see the data that was used to train the tree flow through the tree.

This data is called **training data** because it was used to train the model.



Bag of marbles with replacement



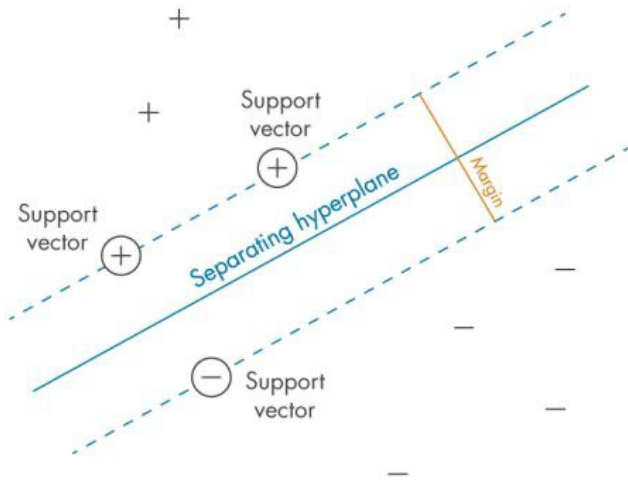


	0	1	2	3	4	type
State	KS	OH	NJ	OH	OK	object
Account length	128	107	137	84	75	int64
Area code	415	415	415	408	415	int64
International plan	No	No	No	Yes	Yes	object
Voice mail plan	Yes	Yes	No	No	No	object
Number vmail messages	25	26	0	0	0	int64
Total day minutes	265.1	161.6	243.4	299.4	166.7	float64
Total day calls	110	123	114	71	113	int64
Total day charge	45.07	27.47	41.38	50.9	28.34	float64
Total eve minutes	197.4	195.5	121.2	61.9	148.3	float64
Total eve calls	99	103	110	88	122	int64
Total eve charge	16.78	16.62	10.3	5.26	12.61	float64
Total night minutes	244.7	254.4	162.6	196.9	186.9	float64
Total night calls	91	103	104	89	121	int64
Total night charge	11.01	11.45	7.32	8.86	8.41	float64
Total intl minutes	10	13.7	12.2	6.6	10.1	float64
Total intl calls	3	3	5	7	3	int64
Total intl charge	2.7	3.7	3.29	1.78	2.73	float64
Customer service calls	1	1	0	2	3	int64
Churn	False	False	False	False	False	bool

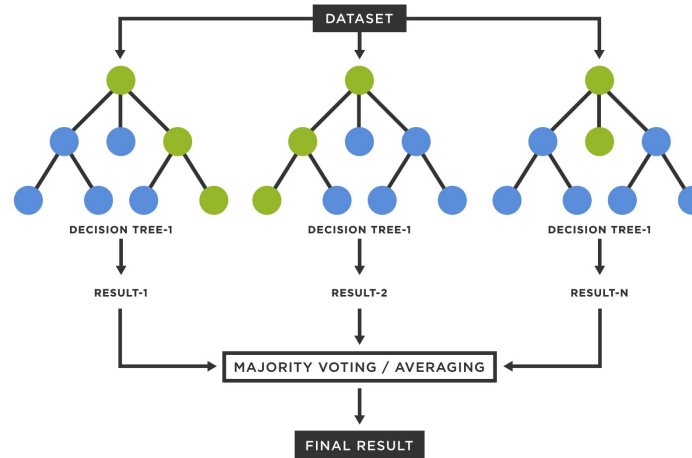
Target variable

3 different approaches to churn prediction

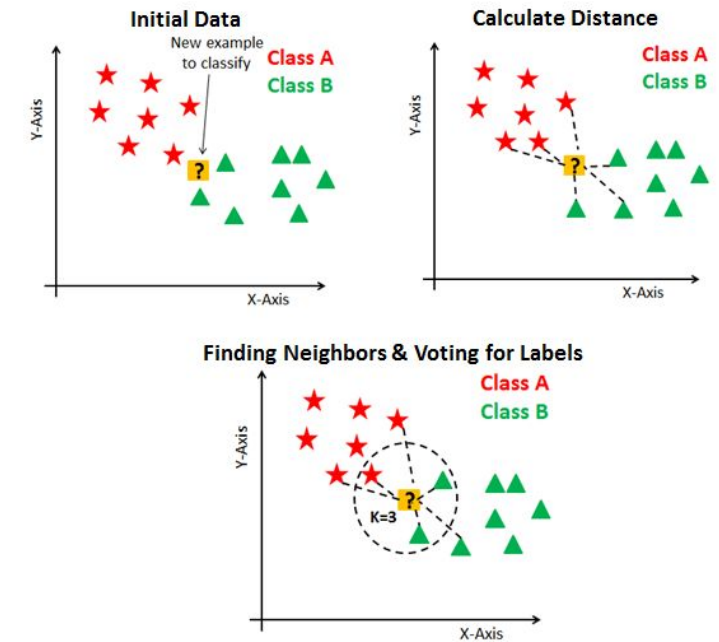
Support vector machine



Random forest



K-nearest neighbors



Confusion Matrix

Actual Classes



Predicted Classes



10	1	2
3	12	3
1	4	15



RATES

$$TP = 10$$

$$TN = (12 + 3 + 4 + 15)$$

$$FP = (1 + 2)$$

$$FN = (3 + 1)$$

Accuracy, recall, precision, F1 score

Accuracy, Recall, Precision



$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} = \frac{\text{What We Got Right}}{\text{All Cases}}$$

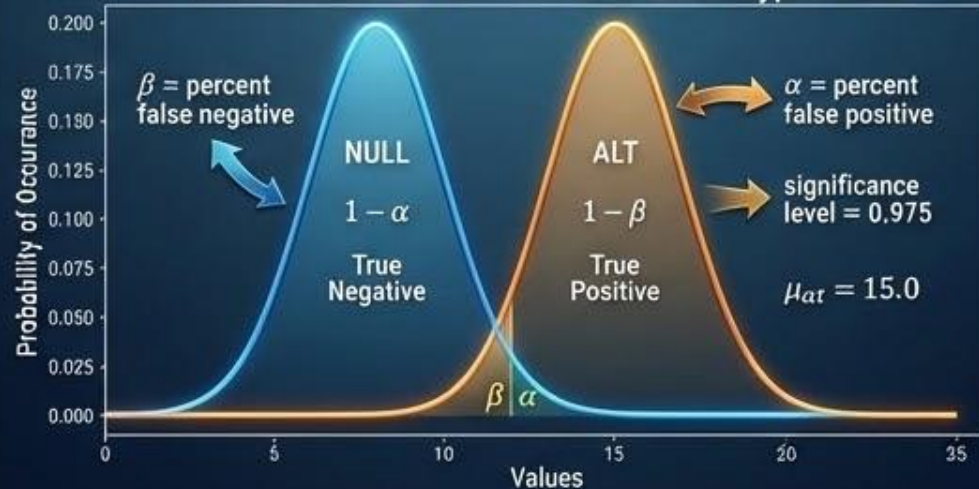


$$\text{Recall} = \frac{TP}{TP + FN} = \frac{\text{What Positives We Got Right}}{\text{All Actual Positives}}$$



$$\text{Precision} = \frac{TP}{TP + FP} = \frac{\text{What Positives We Got Right}}{\text{All Positive Predictions}}$$

Distributions For Null And Alternative Hypotheses

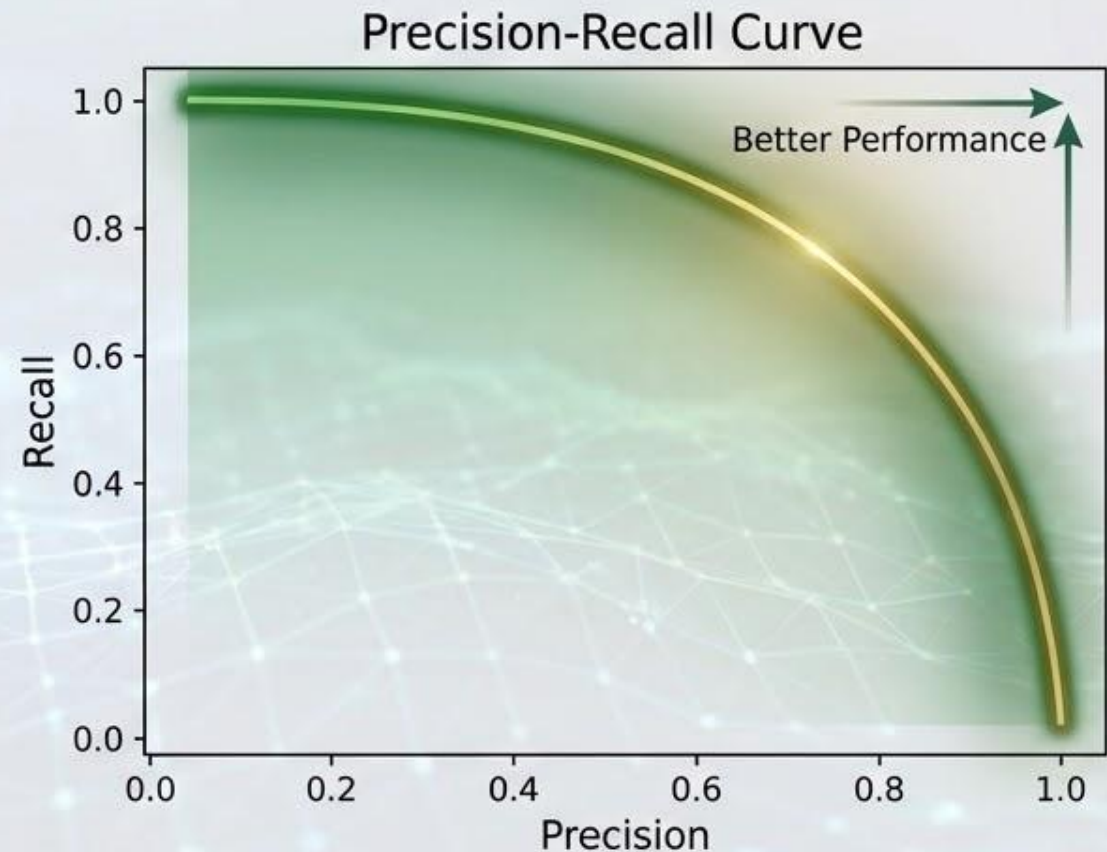
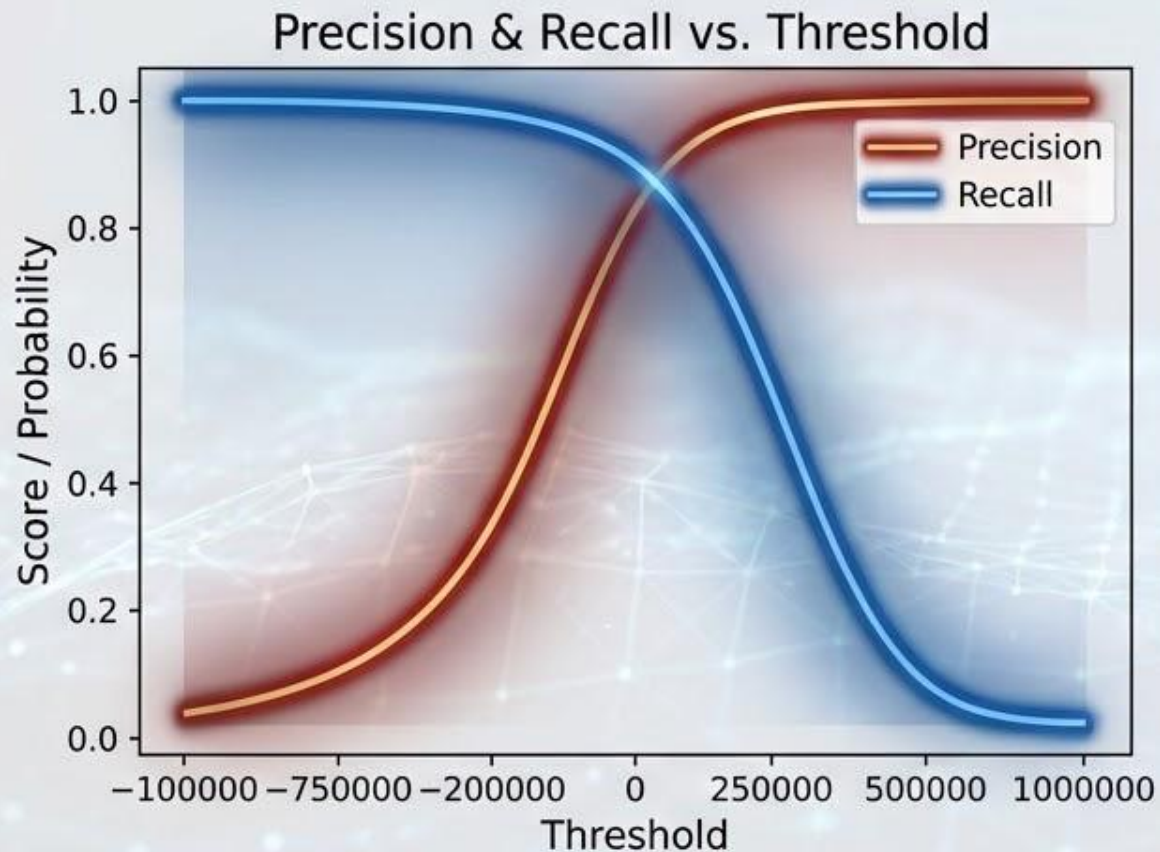


F1 Score

$$F_1 = \frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$$

F_1 = Harmonic Mean Of Recall and Precision

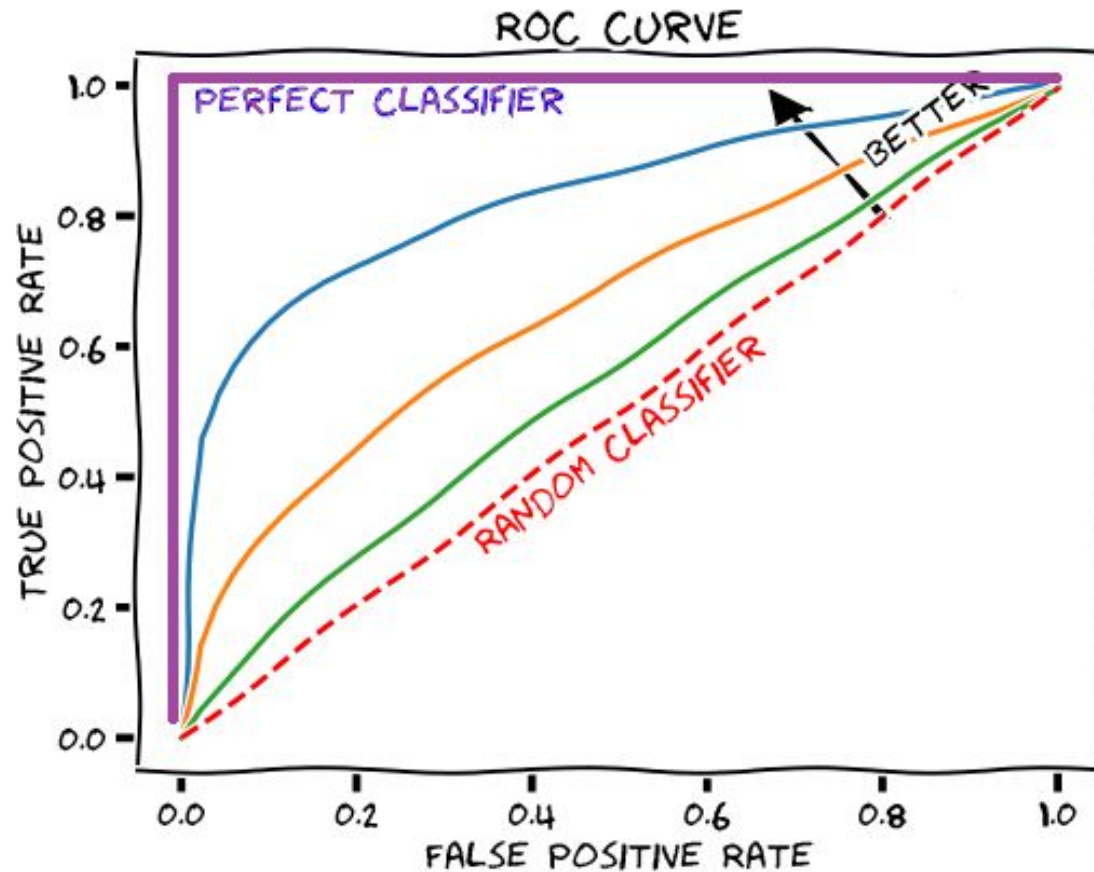
Precision vs. Recall



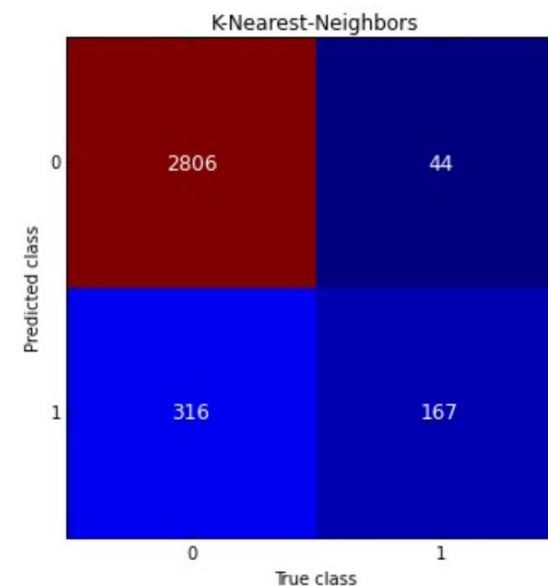
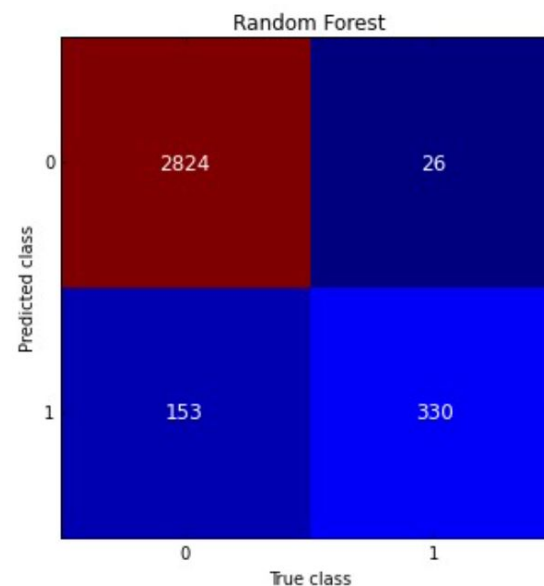
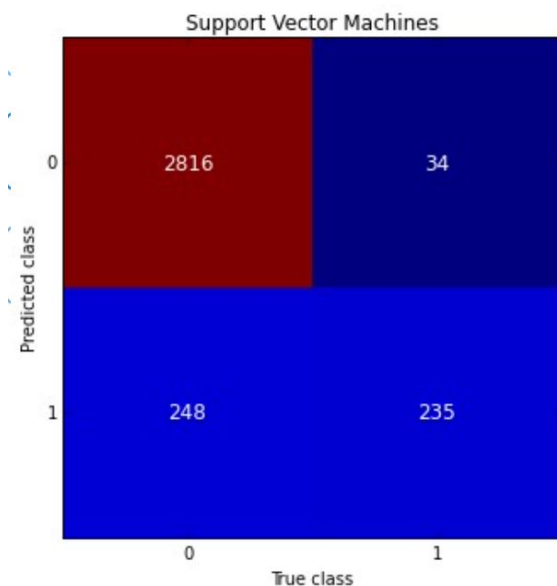
Other model performance metrics (ROC curve) receiver operating characteristics

plots the true positive rate (also called recall)

against the false positive rate (ratio of incorrectly classified negative instances)



Which is the best predictor?

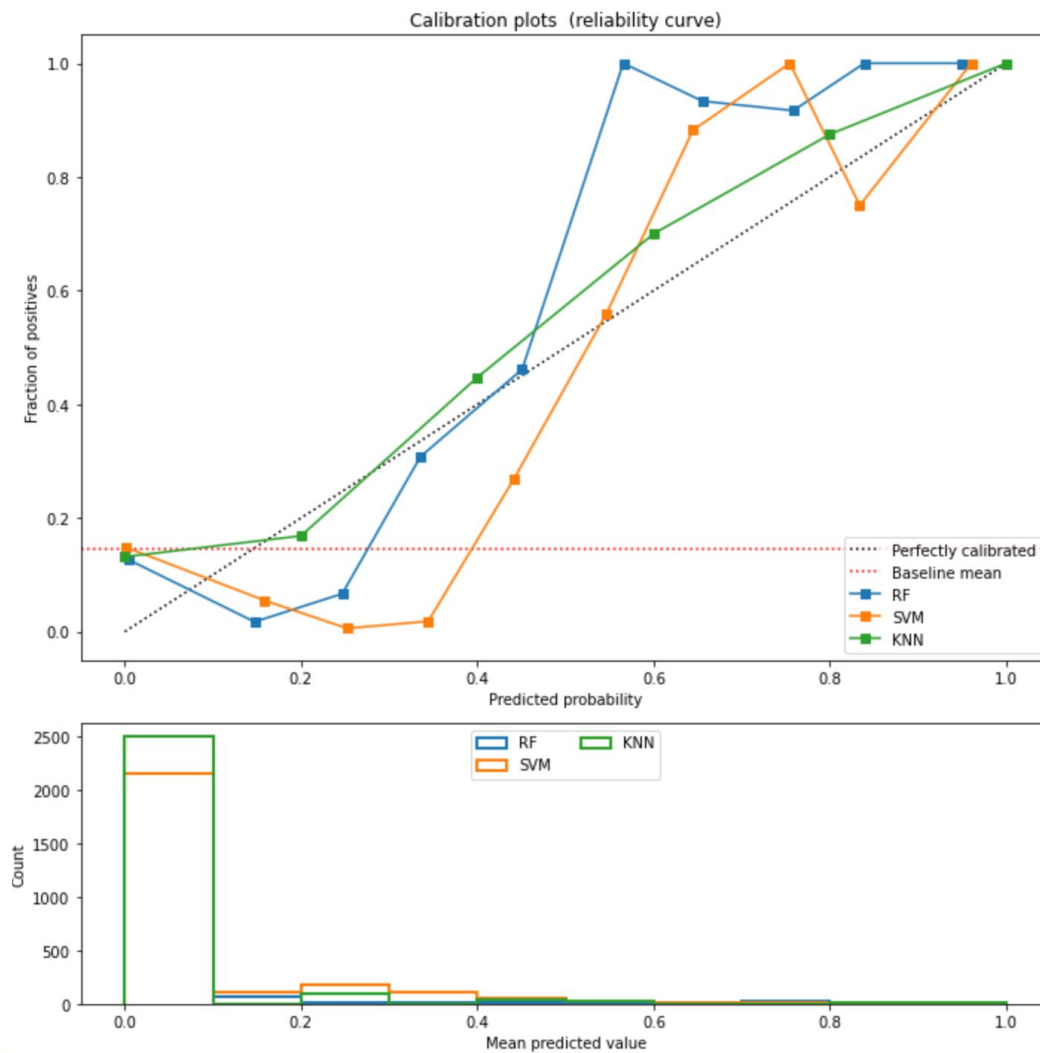


Accuracy Support vector machines:
0.918

Recall ~.5 Random forest: ~.667

Precision ~.87 ~.93 K-nearest-neighbors: ~.8

Which is the best predictor (calibration)?



Note -- Lower calibration is better, higher discrimination is better

Support vector machines:

Calibration Error 0.0227

Discrimination 0.0137

Random forests:

Calibration Error 0.0167

Discrimination 0.0133

K-nearest-neighbors:

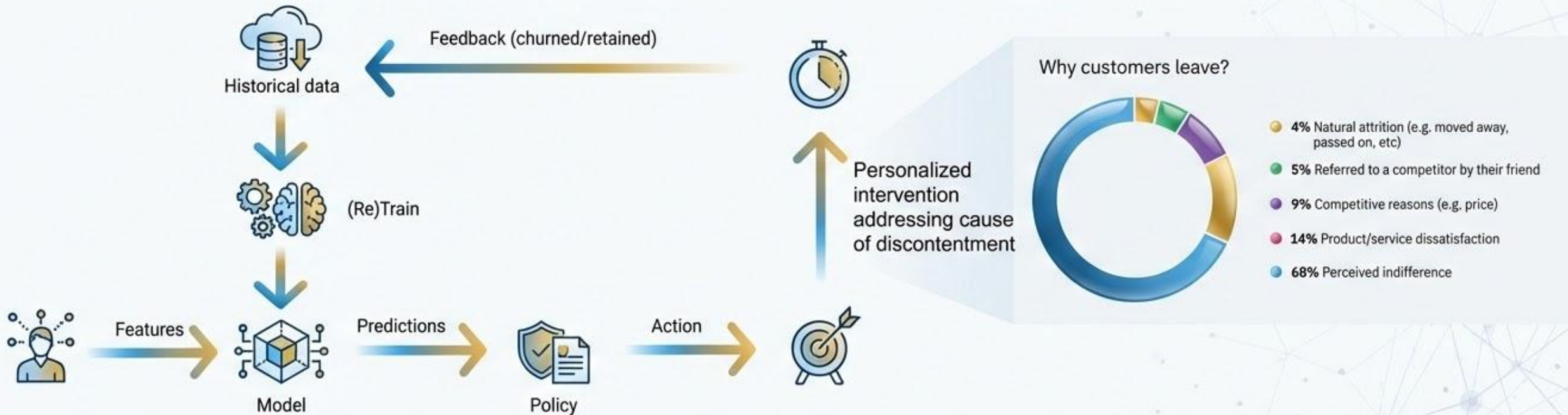
Calibration Error 0.0168

Discrimination 0.0063

Author: John Woolley

Credit for original idea, calibration and discrimination coefficient code: Eric Chiang

ML enhanced operating model



Virtuous cycle of retention, trust, and loyalty



PCA



Visual Inspection

This chart will help you build your mental repertoire for identifying wines by hue and intensity. You will find it useful for blind tasting and assessing quality.

Hue in Red Wine

Pigment in red wine is affected by several factors including pH level. Wines with red-based hues tend to have a lower pH than wines with blue-based hues.

Advice for Viewing Wine

Look at wines under bright, diffused, natural lighting over a white background for best results. Hue is best observed at the point where the wine meets the glass.

WINE FOLLY

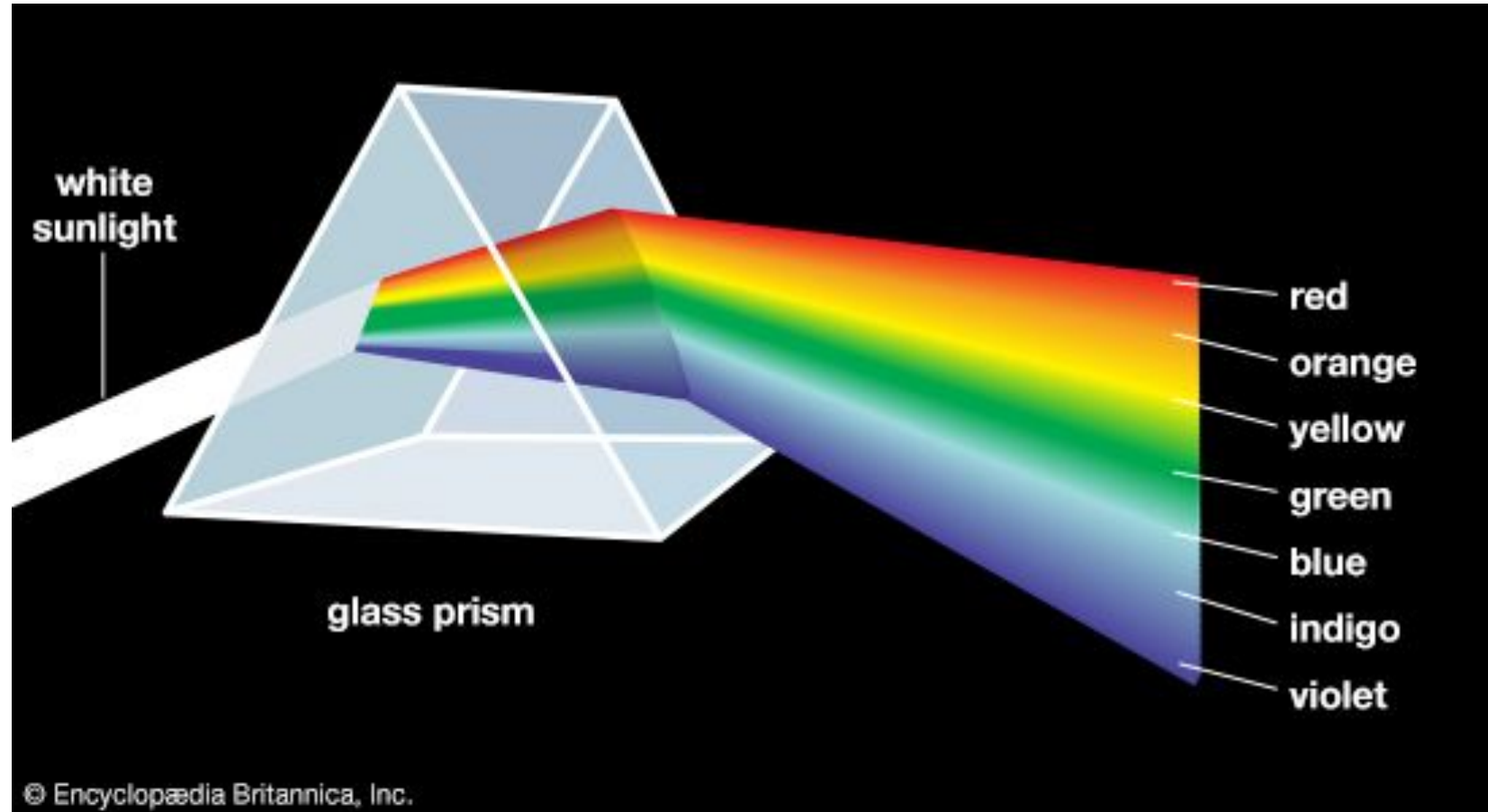
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Made in Seattle, WA, USA
Learn more at winefolly.com

<https://stats.stackexchange.com/questions/2691/making-sense-of-principal-component-analysis-eigenvectors-eigenvalues/2698>

<https://winefolly.com/tips/wine-color-chart/>

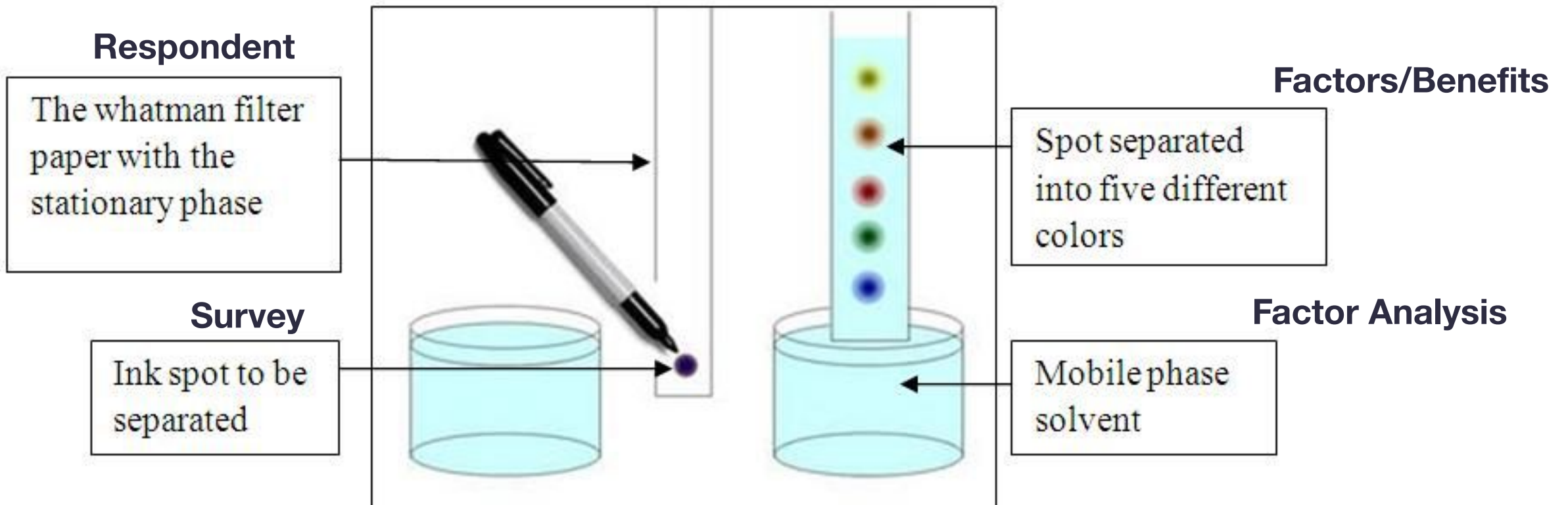
xPRO

Prism



Factor Analysis: A Colorful Analogy

□ Chromatography



Factor Analysis in Product Positioning

- 30 respondents were asked to rate the following statements on a 7-point Scale:

1 = strongly disagree; 7 = strongly agree

1. It is important for my snacks to have low calories
2. Balance of flavors is important in my snacks
3. Snacks should have good nutritious value
4. I prefer snacks that are easy to carry with me
5. High fiber content is not an important quality for my snack choices
6. The most important consideration in buying snacks is taste



Identifying Factors in Snack Consumption

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	2.731	45.520	45.520	2.731	45.520	45.520
2	2.218	36.969	82.488	2.218	36.969	82.488
3	.442	7.360	89.848			
4	.341	5.688	95.536			
5	.183	3.044	98.580			
6	.085	1.420	100.000			

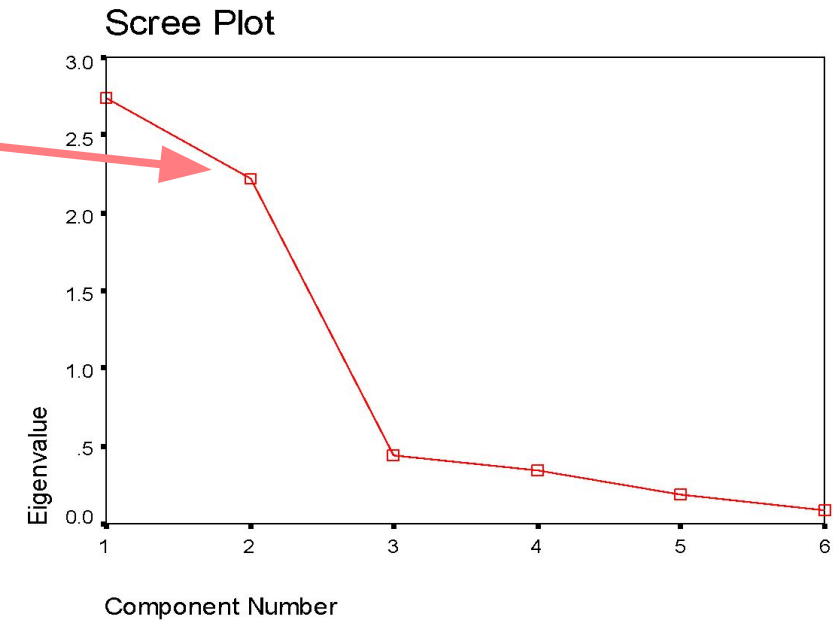
Extraction Method: Principal Component Analysis.

Furthermore, there is very little benefit from an additional third factor.

The top 2 factors account for 82% of the variance

Factor Analysis: Determine the # of Factors

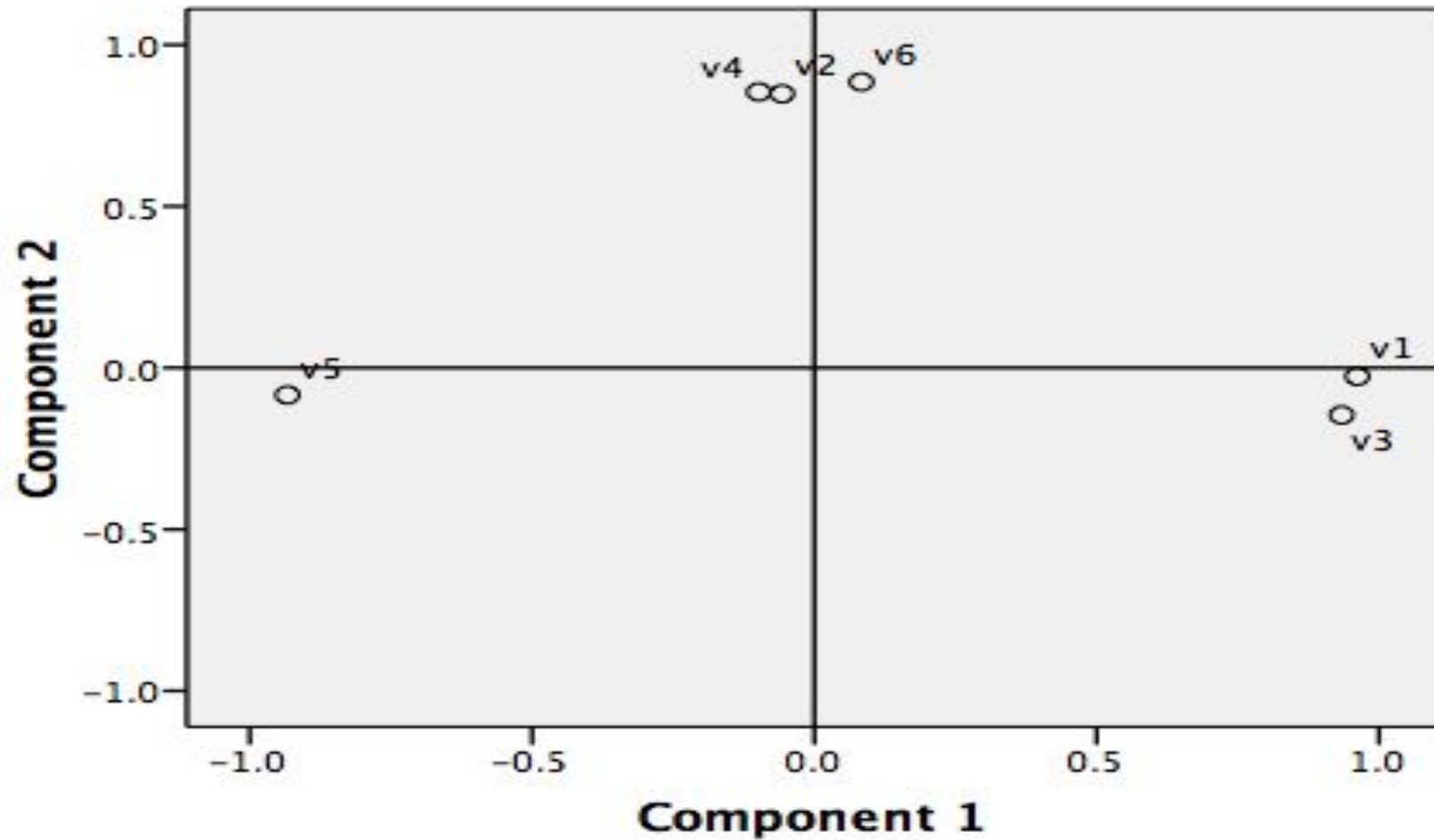
- Two factors in this example
Scree plot “elbow”
Two eigenvalues above 1
Increase in cumulative % explained variance is small when going from 2 to 3 factors
- Explained variance by first two factors
82%, which is excellent



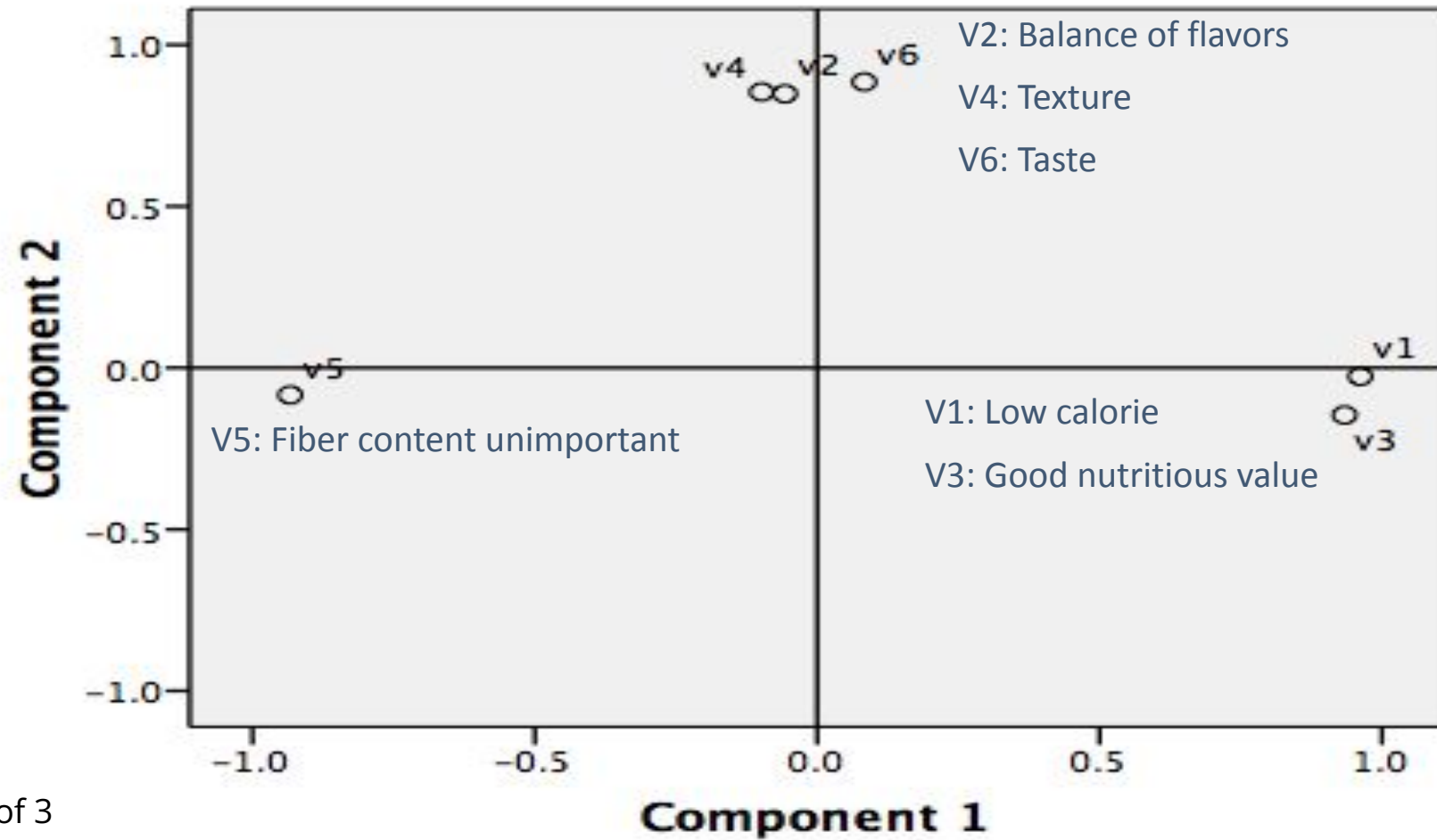
Component Plot in Rotated Space

Rotated Component Matrix

	Component	
	1	2
V1	0.962	-0.027
V2	-0.057	0.848
V3	0.934	-0.146
V4	-0.098	0.854
V5	-0.933	-0.084
V6	0.083	0.885



Component Plot in Rotated Space



best practices to have a minimum of 3 factors per component

Algorithm Acceleration in the AI Centric Operating Model



NOTE: The ultimate business value lies in the speed of the loop—turning strategy into automated execution.

Why ML when I can AI(LLM)?

For an applicant with an annual income of \$50,000, a credit score of 700, 5 years of employment, what is the predicted approved loan amount according to this enhanced model?...

Show thinking

To determine the predicted approved loan amount, we must first reverse-engineer the "Enhanced Model" by analyzing the provided dataset. The data appears to be a linear regression model where the inputs (Income, Credit Score, Years of Employment) predict the Approved Loan Amount.

1. Data Parsing

The raw data provided is a continuous stream of concatenated numbers. By observing standard magnitudes for these variables, we can separate the columns as follows:

- Income: ~\$50,000 (5 decimals)
- Credit Score: ~90-900 (variable decimals, likely 5-7)
- Years: ~19-22 (8 decimals)
- Loan Amount: ~\$13,000-\$15,000 (5 decimals)

Sample Data Points extracted for analysis:

- Row 1: Income: 52,646.08 | Score: 876.63 | Years: 19.26 | Loan: 15,147.70
- Row 2: Income: 50,600.24 | Score: 230.45 | Years: 19.52 | Loan: 13,936.73
- Row 3: Income: 51,468.11 | Score: 245.90 | Years: 22.20 | Loan: 13,828.54
- Row 4: Income: 53,361.34 | Score: 93.88 | Years: 21.31 | Loan: 15,085.66

2. Model Derivation

Using linear regression analysis ($y = \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + C$) on these points, we find the following approximate coefficients:

- β_1 (Annual Income): 0.56
 - Interpretation: For every \$1 of income, loan eligibility increases by roughly \$0.56.
- β_2 (Credit Score): 0.01
 - Interpretation: The credit score in this specific model has a negligible impact (coefficient near zero).
- β_3 (Years of Employment): -221
 - Interpretation: Surprisingly, this model penalizes longer employment tenure, reducing the loan amount by ~\$221 for every year employed.
- Intercept (Constant): -10,086

Formula:

$$Loan \approx 0.56(Income) + 0.01(Score) - 221(Years) - 10,086$$

3. Calculation for the Applicant

Now we apply the derived formula to the applicant's profile:

- Annual Income: \$50,000
- Credit Score: 700
- Years of Employment: 5

$$Loan = 0.56(50,000) + 0.01(700) - 221(5) - 10,086$$

Step-by-step:

- Income Contribution: $0.56 \times 50,000 = 28,000$
- Score Contribution: $0.01 \times 700 = 7$
- Years Deduction: $-221 \times 5 = -1,105$
- Base Subtraction: $-10,086$

Total:

$$28,000 + 7 - 1,105 - 10,086 = 16,816$$

Predicted Result

According to this enhanced model, the predicted approved loan amount is:

\$16,816

Python/scikit

```
1 import pandas as pd
2
3 # Create a DataFrame for the new applicant's data
4 new_applicant_data = pd.DataFrame({
5     'Annual Income': [50000],
6     'Credit Score': [700],
7     'Years of Employment': [5]
8 })
9
10 # Predict the approved loan amount using the multivariable model
11 predicted_loan_amount = multi_model.predict(new_applicant_data)
12
13 print(f"The predicted approved loan amount for the new applicant is: ${predicted_loan_amount[0]:,.2f}")
```

The predicted approved loan amount for the new applicant is: \$14,186.82

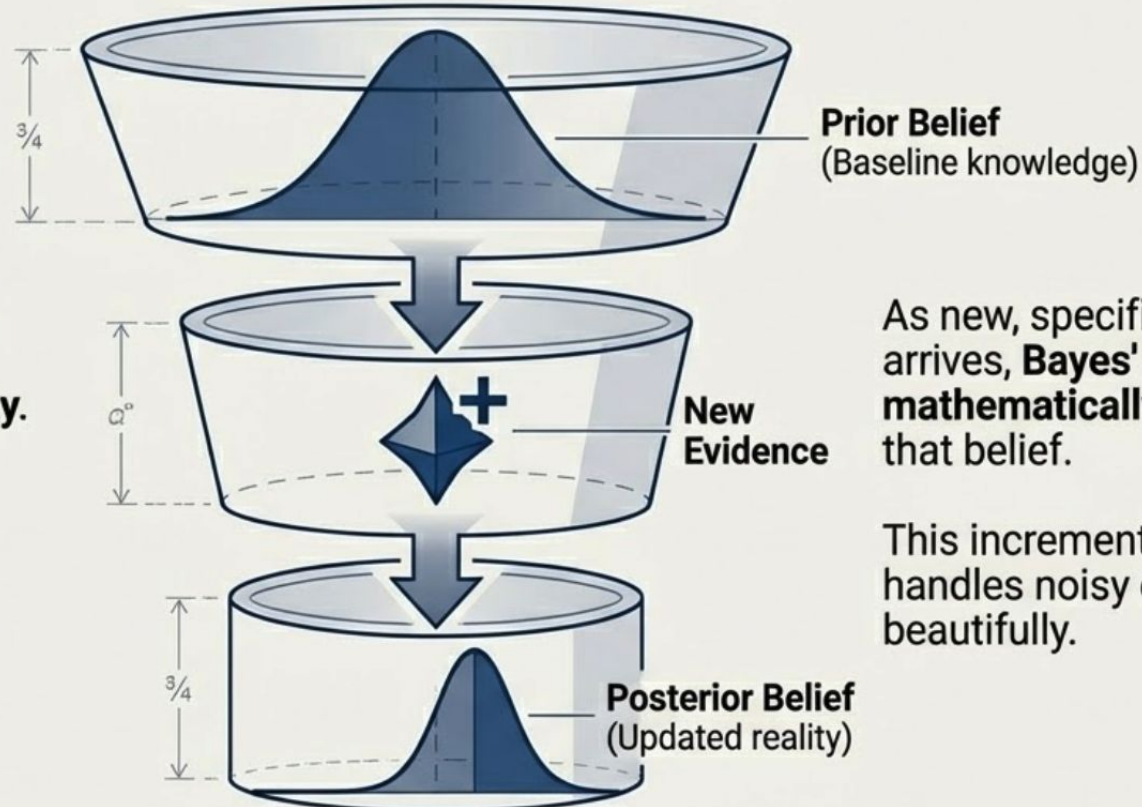
Bayesian Learning Model

Bayesian Models Update Beliefs with Evidence

⚙️ Probabilistic Learning

Bayesian methods do not just guess an output; they **explicitly model uncertainty**.

They start with a **baseline belief** based on domain knowledge.



As new, specific evidence arrives, **Bayes' rule mathematically updates** that belief.

This incremental learning handles noisy data beautifully.

Bayesian Learning Model

Bayesian Application: Cybersecurity Threat Detection

⚙️ Probabilistic Learning

The Prior Belief



Baseline assumption: An employee logging into the corporate network is legitimate (99% probability).

The New Evidence



Observation: The login occurs at 3:00 AM from an IP address in a foreign country where the company has no offices.

The Posterior Belief



The Bayesian algorithm calculates the joint probability of this rare event. The belief in legitimacy instantly drops to 5%, automatically triggering an account lockdown.

Pastry quality & Cancer detection

ANNALS OF TECHNOLOGY

THE PASTRY A.I. THAT LEARNED TO FIGHT CANCER

In Japan, a system designed to distinguish croissants from bear claws has turned out to be capable of a whole lot more.

By James Somers

March 18, 2021

<https://www.newyorker.com/tech/annals-of-technology/the-pastry-ai-that-learned-to-fight-cancer>

Cyto-AiSCAN

BakeryScan was able to identify a cancerous cell under a microscope with 98% accuracy. At first, it was only able to look at one cell at a time. After some modifications and testing, the AI was able to look at an entire slide of cells under a microscope and highlight which cells were cancerous, again, with a 98% accuracy. When treating cancer, the survival rate greatly increases the earlier the cancer is detected. Using the AI dramatically increased the rate that lab technicians could sort through samples and detect the presence of cancerous cells. Brain Co. gave the modified BakerScan a new name: “Cyto-AiSCAN” and it is currently being used and trained by doctors at two major hospitals in Kobe and Kyoto, Japan.

When I heard this story I immediately thought that they must have significantly changed the way that BakerScan operated in order to detect cancer cells. Reporter James Somers for the New Yorker also thought the same thing. He asked the CEO of Brain Co., Hisashi Kambe, the same question.

I asked Kambe how it worked — did it use deep learning? “Original way,” he said. Then, with a huge smile, “Same as bread.”

Nope, same as bread.

<https://towardsdatascience.com/bakeryscan-and-cyto-aiscan-52475b3cb779>

Pastry quality & Cancer detection





Q&A